Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years shipping cloud applications end to end, from **REST** APIs, **SQL**, and React front ends to CI/CD, **Docker**, and **Kubernetes** on **AWS**, with tested code, production support, and security scanning.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Measured and fixed a **performance bottleneck** in multi-tenant SQL dashboards, redesigning Azure SQL and PostgreSQL indexes and access patterns to cut latency from 1.2s to 350ms for 500+ users.
- Built FastAPI services with background workers, explicit job-state transitions, and failure isolation, processing 1K+ jobs at 3x throughput while integrations held 99%+ **production reliability** via CI/CD releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped features **end to end** across React views, Flask REST APIs, MySQL on AWS RDS, and GitHub Actions CI/CD to Render and Vercel, serving 150+ instructors and 10K+ daily requests.
- Designed the frontend-to-backend **workflow automation** for uploads, rubric generation, and grading with concurrent batches of 1K+ uploaded files and fault-tolerant execution, cutting average feedback turnaround by 60% for instructors.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Wrote 20+ **unit and integration test** suites with PyTest and Mockito in nightly Jenkins pipelines that built, installed, and validated releases across Linux, Windows, and macOS with coverage reports.
- Optimized large-scale API integrations with indexing, caching, and pagination, improving response times 30%, and led an AWS Glue and S3 ETL redesign that boosted **pipeline efficiency** 68%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight TypeScript and Node.js **microservices** with React for resource allocation, billing, invoices, and email and SMS notifications, deployed as Docker builds to Kubernetes on AWS and syncing 2M+ records.
- Secured CI/CD with JWT, role-based access, and Snyk and OWASP **security scanning**, gating staged releases with approvals and rolling zero-downtime updates at 95%+ build validation, monitored with Sentry.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **booking management** APIs for a home-services platform covering time-slot reservations, pricing, coupons, and payment status with Firebase real-time updates and transactional locking, sustaining sub-200ms latency across 50K+ records.
- Designed the end-to-end architecture with 12+ **integration** points and presented it to the client, then automated refund and dispute handling through idempotent webhooks, cutting manual work 70%.

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, HTML, CSS
Full Stack: React, Next.js, Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Webhooks, Microservices
Data: PostgreSQL, MySQL, SQL Server, MongoDB, Firebase, Query Optimization, Indexing
Cloud & DevOps: AWS (EC2, S3, RDS, ECS/Fargate, Glue), Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Jenkins
Quality & Security: Unit Testing, Integration Testing, PyTest, Jest, Selenium, JWT, RBAC, Snyk, OWASP, Sentry
Practices: Technical Documentation, Architecture Diagrams, Production Support, Code Reviews, Agile

Projects

Order and Inventory Management Service | *TypeScript, Node.js, React, PostgreSQL, RabbitMQ, Docker, Kubernetes*

- Designed an order and inventory service that ingests orders via REST, reserves stock with transactional SQL, publishes pick and ship events through **RabbitMQ**, and tracks order status in a React **dashboard**.

FinNexus Live | *Kubernetes, Next.js, TypeScript, MongoDB, WebSockets*

- Built and deployed a full-stack **Kubernetes** fintech platform in Next.js and TypeScript with a token-based firewall, Plaid integration, and WebSocket updates across 1M+ transactions for 50K+ users.

Linguistix | *Python, Docker, AWS ECS/Fargate, CodePipeline, CI/CD*

- Containerized with Docker and deployed a Python REST data processing API on **AWS** ECS/Fargate with automated CodePipeline CI/CD releases and validation checks, achieving 87% accuracy and production-grade reliability.